

# Fluid Dynamics of the Atmosphere and Ocean

## Problem Set U1

Unassessed.

1. In a two-layer shallow water system in geostrophic balance, show that the difference in the velocity between the two layers is proportional to the slope of the interface between the layers.
2. Show that the vertical velocity within a shallow water system is given by

$$w = \frac{z - \eta_b}{h} \frac{Dh}{Dt} + \frac{D\eta_b}{Dt}. \quad (1)$$

Interpret this result, showing that it gives sensible answers at the top and bottom of the fluid layer. You may do the problem with  $\eta_b = 0$ .

3. Derive the dispersion relation for the linearized, rotating, shallow water equations. The equations are cubic, explain why there are only two frequencies. Plot the dispersion relation using software of your choice (e.g., Python, Matlab) or, if you must, by hand. Using reasonable values of your choice for the various parameters, estimate approximate values of the frequencies of these waves (known as Poincaré waves) for the atmosphere and ocean.
4. *Geostrophic adjustment with a velocity jump.*

- (a) Show that the *linearized* potential vorticity,  $q'$ , for the shallow water system is given by

$$q' = \zeta' - f_0 \frac{h'}{H},$$

using standard notation. {5}

- (b) If the flow is in geostrophic balance show that the relative vorticity is given by

$$\zeta' = \nabla^2 \psi.$$

where  $\psi = gh'/f_0$ . Hence show that the potential vorticity is then given by

$$q' = \nabla^2 \psi - \frac{1}{L_d^2} \psi,$$

and write down an expression for  $L_d$ . (If you wish you may take  $g = 1$ .) {5}

- (c) Suppose that the initial flow has a flat surface with  $u' = 0$  everywhere and Suppose that initially the fluid surface is flat, the zonal velocity ( $u$ ) is zero and the meridional velocity is given by

$$v(x) = v_0 \operatorname{sgn}(x)$$

- (i) Find the equilibrium height and velocity fields at  $t = \infty$ , in the linear approximation.

- (ii) What are the initial and final kinetic and potential energies?

*Hint:* The potential vorticity is  $q = \zeta - f_0 \eta / H$ , so that the initial potential vorticity is given by

$$q = 2v_0 \delta(x).$$

where  $\delta(x)$  is the Dirac delta function.

{10}

{20}